

Dexter Fernandes

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AI Engineer

AI Engineer with experience building end-to-end production AI systems in Python across training, deployment, observability, and backend-integrated workflows. Background in shipping applied AI under real-world constraints, combining model development, service design, and production engineering to deliver robust user-facing systems for clients such as British Petroleum and Austin Industries. Strong track record of turning ambiguous product goals into practical systems for search, extraction, and workflow automation, backed by solid experimentation and model evaluation.

PROJECTS

Counsel Copilot

Feb 2026 - Present

- Building Counsel Copilot, a Python-based prototype for legal document editing, retrieval, and research, designed to support contract review workflows.
- Implementing a FastAPI-backed service layer with OpenAI models, PageIndex-powered RAG, and python-docx document handling to generate grounded answers, propose structured edits, and preserve DOCX formatting integrity.
- Designing a human-in-the-loop redlining workflow that proposes structured edits, previews document changes, and requires explicit approval before updates are applied.
- Developing a citation-grounded legal research flow that combines query interpretation, document retrieval, and structured answer generation to improve traceability and user trust.
- Creating a DOCX processing pipeline for section-aware parsing, controlled edit generation, and reliable export of revised legal drafts while preserving document structure.
- Targeting retrieval hit rate@5 above 85%, citation support rate above 95%, and faithfulness above 0.90 on a held-out evaluation set built from CUAD - a contract documents dataset.

End-to-End Damage Severity & Claim Assessment

- Developed an end-to-end automotive claims assessment pipeline combining computer vision-based damage detection with a trained severity-classification model, achieving accurate triage for real-world vehicle damage scenarios.
- Built a lightweight repair-insight engine that transforms model outputs into insurer-friendly explanations, using structured reasoning + LLM-generated narratives to deliver clear, interpretable claim summaries.
- Designed the entire workflow (data preprocessing, model training, evaluation, and a user-facing inference module) demonstrating practical understanding of insurance claims processes.

WORK EXPERIENCE

Esbaa • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Engineered production AI systems for 24/7 industrial safety analytics in Python, deploying real-time perception pipelines on edge hardware and improving F1 score by 15% under harsh field conditions.
- Built end-to-end data ingestion, cleansing, training, and validation pipelines for multimodal sensor datasets using MLflow, DVC, and Optuna, cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, improving fault isolation, increasing MTBF by 35%, and strengthening service continuity through logging, health checks, retries, and safe restarts.
- Developed an internal web dashboard in Django to surface inference outputs, system health, and performance metrics, enabling faster debugging and root-cause analysis across deployed systems.

- Led validation, benchmarking, and observability for fielded AI deployments, reducing crash frequency by 67% and documenting failure modes, recovery procedures, and deployment guardrails for engineers operating live systems.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.
- Documented deployment guardrails, failure modes, and recovery procedures, reducing cognitive load for engineers operating and maintaining fielded systems.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI

- Led a 4-member team to design, train, and deploy end-to-end production AI models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices.
- Built automated labelling, dataset versioning, and performance-tracking pipelines, repurposing detection models for annotation and cutting manual labelling effort by 80%, accelerating iteration cycles in modern Python.
- Collaborated with hardware, software, and QA teams to integrate AI modules into production systems, mentoring engineers on optimisation and deployment practices to improve delivery consistency.
- Mentored engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.
- Presented key findings and metrics to leadership and clients, translating complex performance results into actionable insights for deployment and further optimisation.
- Designed data handling and retention practices aligned with GDPR principles, including data minimisation and controlled access, within production computer vision systems.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Designed and implemented a real-time SLAM system using RTAB-Map, integrating multimodal LiDAR and radiological data for localisation and mapping in inspection scenarios leveraging the Clearpath Husky development platform.
- Evaluated RTAB-Map (LiDAR-only), LIO-SAM, and Cartographer across comparable scenarios to assess mapping and localisation behaviour in simulation prior to real world deployment.
- Containerised the full system using Docker to ensure reproducibility, version control, and cross-platform deployment.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Strong foundation in signals, systems, embedded hardware, and control, supporting later work in real-time AI and perception systems.

SKILLS

- **Large Language Models:** Citation-grounded response generation, Error analysis, Experimentation and eval, Information extraction, Model fine-tuning, Prompt engineering, Retrieval-augmented generation (RAG), Search and Retrieval, Workflow automation
- **Cloud, Edge & MLOps:** Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Intel VPUs, Microsoft Azure, MLFlow, Object storage, Service-layer orchestration
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Linux, NumPy, Pandas, PG vector, PostgreSQL, Python, Qt
- **Deep Learning and Optimisation:** Bayesian Optimisation, ONNX, OpenVINO, PyTorch, TensorFlow, TensorRT

- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership